Assignment#3:
This assignment combines the unsteady blade loads with the dynamic structural response of a wind turbine rotor. Consider a simple structural model for a wind turbine as sketched in Figure 1. Put cone and tilt angle to 0° and you may also neglect the effect of the tower on the inflow. Dynamic stall, dynamic wake and your controller should be on by default as well as gravitational loads on the blades.

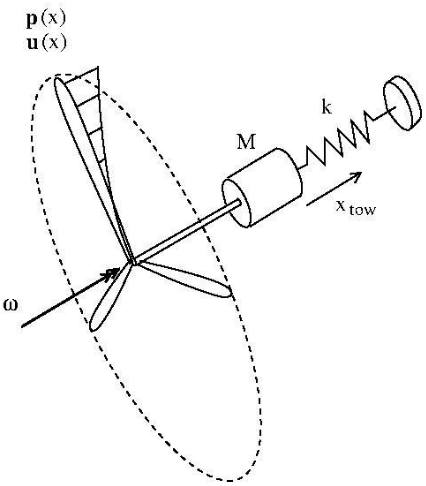

Figure 1: *A simple structural model of a wind turbine rotor. M is the mass of the nacelle, drivetrain and generator. The spring with stiffness, k, models the elastic deformation of the tower.*

The tower stiffness (in and out of the wind, z-direction) is modelled as a simple spring with a spring stiffness, $k=1.7\cdot 10^6$N/m and the mass of the nacelle including gearbox and generator is M=446000kg.
The blade deflections are modelled using a modal approach as a linear combination of the three lowest blade eigenmodes (1f = first flap, 1e first edge and 2f=second flap) as

$$\underline{u}_n(r,t)=\begin{pmatrix} u_y(r) \\ u_z(r) \end{pmatrix}=q_n^1(t)\begin{pmatrix} u_y^{1f} \\ u_z^{1f} \end{pmatrix}+q_n^2(t)\begin{pmatrix} u_y^{1e} \\ u_z^{1e} \end{pmatrix}+ q_n^3(t)\begin{pmatrix} u_y^{2f} \\ u_z^{2f} \end{pmatrix}$$ n denotes the blade number 1,2 or3

There are 3 elastic DOFs per blade (q_n^1, q_n^2, q_n^3 n=1,2,3). The mode shapes ($u_{z,0}$ and $u_{y,0}$) are given for 0 pitch angle, so remember to rotate them in case of a non-zero pitch and update the mass and stiffness matrix to calculate the deflection in the rotor coordinate system. The subscript (k=1,2 or 3) here indicates the mode shape

$$u_{y,R}^k = u_{z,0}^k \sin\theta_p + u_{y,0}^k \cos\theta_p$$
$$u_{z,R}^k = u_{z,0}^k \cos\theta_p - u_{y,0}^k \sin\theta_p$$

The 11 DOFs are

$$\mathbf{GX}=(z,\theta,q_1^1,q_1^2,q_1^3,q_2^1,q_2^2,q_2^3,q_3^1,q_3^2,q_3^3)^T$$

The shaft is considered stiff in torsion.

If your code becomes unstable, you may try and stabilize it by including some small structural damping (see Eq. 12.39 in Aerodynamics of wind turbines book) and using around 3% logarithmic decrement damping value.

Q#1: Solve first a 5 DOF system, with only one elastic blade

$$\mathbf{GX} = (z, \theta, q_1^1, q_1^2, q_1^3)^T$$

-Show how to calculate the relative wind speeds for the blade elements along the blades.

-Calculate and plot for a turbulent wind speed Vo=7m/s and 18m/s in time:

1) blade tip deflection in flapwise and edgewise direction
2) the tower displacement.
3) Root bending moments at r=2.8 m.
4) Also provide PSD plots and explain.

Note that the equation for computing the rotational speed is a direct part of the EOMs.

Q#2 Calculate the blade deflections for a steady wind of Vo=7 m/s having a wind shear exponent of 0.2. Compare the PSDs with the turbulent case.

Q#3: Now consider the full 11DOF system where the elastic deflections of the two remaining blades are also included.

-Derive the mass and stiffness matrices and the generalized force vector for this system.

-Solve the full eigenvalue problem $(\mathbf{K}^{-1})\mathbf{M}\cdot\mathbf{GX} = \frac{1}{\omega^2}\mathbf{GX}$ to determine the natural frequencies and mode shapes for the coupled system. Discuss the results. To solve the eigenvalue problem you may have to remove the equation for azimuthal position, θ, since there is no stiffness in rotation. Or alternatively add a stiffness in rotation.

-Simulate and document in time and frequency domain a wind speed Vo=7m/s and 18m/s with and without turbulence. Make the simulation as realistic as possible, so include all your models, perhaps also tower shadow.

- Compare the tip deflection with ASHES for the cases without turbulence.

Try not to exceed more than 10 pages.